



Summative Assessment 2

**A Simulation and Calibration of the Solow–Swan Neoclassical Growth Model:
Analyzing Parameter Sensitivity on the Philippine Economic Trajectory**

Far Eastern University - Manila

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**Submitted to:
Mrs. Nikki Blas-Olay
Professor**

**Submitted by:
Baybayon, Darlyn Antoinette
Bazar, Lindsay Faith
Mayol, Jose Raphael
Morillo, Jade Marco**

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I. Introduction

A. Background of the Study

Economic growth has long been a central concern of policymakers in the Philippines, particularly because the country has consistently lagged behind many of its Asian neighbors despite starting from comparable levels of development in the mid-20th century. Between the 1960s and the late 1980s, Asian economies such as South Korea experienced rapid improvements in per capita income, while the Philippines exhibited slow, volatile, and often stagnant growth. This divergence reflects bigger structural differences in development strategies across the region.

Many East Asian economies achieved rapid and sustained growth through a state-guided development model that strategically combined export-oriented industrialization with strong investment in human capital (Wong, 2025). Governments played an active role in nurturing domestic industries by directing credit, protecting new sectors through industrial policy, and promoting technological adoption to support manufacturing exports. This coordinated approach, often described as the “flying geese” model, was anchored in high savings rates, strong educational systems, and a carefully sequenced opening to foreign investment and global markets. These factors created a foundation for rapid productivity gains and structural transformation, enabling countries like South Korea, Taiwan, and Singapore to transition quickly from low-income to high-income status.

The Philippines did not follow an equivalent strategic path. Its historically weaker growth performance relative to its East Asian neighbors has been linked to lower degrees of trade openness and more volatile investment rates (Kongsamut & Vamvakidis, 2000). However, the shift toward more outward-oriented and market-based policies after 1986 laid the groundwork for improved economic resilience. This transition contributed to the relatively mild contraction experienced by the Philippines during the Asian financial crisis. Increased openness and the rise in foreign direct investment are identified as important factors behind the more optimistic economic performance observed in the years that followed.

Despite these policy improvements, the Philippines continued to face structural challenges. Low investment rates, rapid population growth, political uncertainty, and inconsistent human capital formation limited the country's ability to match the rapid productivity gains achieved by its regional peers (Canlas & Cham, 2008). These issues contributed to persistent underperformance and prevented the Philippine economy from fully realizing the benefits observed in countries that implemented more aggressive investment and industrialization strategies.

In recent years, the Philippine economy has continued to demonstrate steady growth, driven largely by strong domestic consumption. Despite this positive trend, the country faces persistent challenges, including elevated inflation, an enduring infrastructure deficit, and subdued levels of investment. According to recent assessments, these structural headwinds require policy shifts

aimed at promoting long-term and sustainable development rather than short-term economic resilience (Council on Foreign Relations, 2025). Understanding how these modern challenges connect to historical patterns of weak capital accumulation, population pressures, and human capital constraints remains essential for assessing the country's prospects for sustained growth.

To explain both historical and contemporary growth trends, Canlas (2003) applied the Solow–Swan Neoclassical Growth Model, which is one of the foundational frameworks in macroeconomic growth theory. The model analyzes how savings, population growth, physical capital, and human capital accumulation influence long-run economic outcomes. It also provides a link between mathematical growth theory and empirical data by assessing whether the Philippine growth trajectory aligns with theoretical predictions. Building on this work, our report uses updated datasets and additional socio-economic variables to capture the evolving dynamics of the Philippine economy.

B. Real-World Problem Addressed

The study examines the persistent gap between the Philippines and its East Asian neighbors in terms of long-term economic growth and development. Although the Philippines began the postwar period with levels of income and industrial potential comparable to those of countries such as South Korea and Malaysia, it struggled to sustain high and stable growth. Research highlights that many East Asian economies pursued coordinated, state-guided development strategies that combined export-oriented industrialization with strong investment in human capital, which enabled them to transition rapidly into high-productivity

manufacturing (Wong, 2025). The absence of a comparable strategic approach in the Philippines contributed to an early divergence in economic performance.

One major challenge relates to the country's low and volatile investment rates, which have limited the accumulation of physical capital and slowed productivity growth. Several studies note that the Philippines maintained lower degrees of trade openness and less stable investment conditions compared to its regional neighbors, which limited its ability to take advantage of global chains and foreign market opportunities (Kongsamut & Vamvakidis, 2000). At the same time, rapid population growth placed additional pressure on households and public resources, reducing savings and diluting capital per worker. These weakened the foundations for sustained capital deepening and long-term productivity improvements.

While school enrollment levels improved over time, research shows that the quality of education and learning outcomes remained inconsistent, limiting the capacity of Filipino workers to shift into higher-value-added sectors (Canlas & Cham, 2008). These gaps hindered technological adoption and reduced the country's ability to match the pace experienced by neighboring economies.

Institutional conditions further shaped these outcomes. High dependency ratios, governance challenges, and weaknesses in policy implementation limited the effectiveness of economic reforms in the late twentieth century. Although outward-oriented and market-based policies implemented after 1986 strengthened economic resilience and contributed to a more stable growth environment, particularly during episodes such as the Asian financial crisis, these reforms did

not fully resolve long-standing structural barriers (Council on Foreign Relations, 2025).

Understanding how these structural, demographic, and institutional factors interact is essential for explaining why the Philippines has not achieved the sustained growth experienced by its regional peers. It also provides the basis for examining how savings behavior, population dynamics, investment patterns, and human capital development continue to shape the country's long-term economic prospects (Canlas, 2003).

C. Purpose of the Report

The report aims to empirically analyze the long-term economic growth performance of the Philippines relative to its Asian counterparts by applying the Extended Solow–Swan Neoclassical Growth Model. This study specifically seeks to update and re-evaluate the foundational work of Canlas (2003) by employing an extended and contemporary dataset, spanning 1981 to 2021. The initial objective is to test whether the core theoretical predictions of the Solow framework regarding the effects of physical capital accumulation, population growth, and human capital remain statistically valid and significant drivers of Philippine output per worker. To achieve this, the analysis first involves performing Ordinary Least Squares estimations and Steady-State Simulations under various parameter assumptions to determine the empirical consistency of the key growth factors and to graphically compare the model's theoretical growth path with the country's actual, volatile economic trajectory.

Another objective of this report is to identify and address the limitations of the standard Solow model when applied to the structural complexities of the Philippine economy, particularly the challenges associated with using noisy or inadequate proxy variables, such as the Secondary Enrollment Rate for human capital. To overcome these deficiencies, the study proposes and simulates augmented model extensions. These modifications incorporate more refined socio-economic variables, including Tertiary Enrollment as an improved human capital proxy, and the inclusion of Institutional Quality, using the World Governance Indicators, and Demographic Structure, using the Youth Dependency Ratio. The ultimate goal of these extensions is to achieve a superior empirical fit, enhance the model's explanatory power, and consequently provide a more robust and comprehensive quantitative basis for long-term policy recommendations aimed at securing sustained and stable economic development in the Philippines.

II. Description of the Mathematical Model

A. Overview of the Model

This study is anchored in the Extended Solow–Swan Neoclassical Growth Model, adopting the theoretical structure introduced by Mankiw, Romer, and Weil (1992) and previously applied to the Philippine context by Canlas (2003). The model serves as the foundational framework for analyzing the long-run determinants of per capita income growth by moving beyond the standard focus on only physical capital and labor. It explicitly incorporates the accumulation of a critical factor, human capital. The core of the model is a Cobb–Douglas aggregate production function that assumes constant returns to scale for all three factors, but

diminishing returns to each factor individually. The function is expressed as $Y(t) = A(t)K(t)^\alpha H(t)^\beta L(t)^{1-\alpha-\beta}$, $0 < \alpha + \beta < 1$, where $Y(t)$ is total output, $A(t)$ is the level of technology, and $K(t)$, $H(t)$, and $L(t)$ are physical capital, human capital, and labor, respectively. The exponents α , β , and $1 - \alpha - \beta$ represent the output shares paid to the respective factors, with the theoretical constraint that $\alpha + \beta < 1$. This formulation allows for an assessment of how distinct policy levers, specifically the savings rate for physical capital, the investment rate in human capital, and the rate of population or labor force growth, interact to determine the long-run equilibrium. The model's principal utility lies in its ability to predict a convergence toward a steady-state equilibrium, where the accumulation of capital balances out population growth, technological progress, and depreciation. The dynamics of capital accumulation are described by the differential equations for both physical and human capital $\frac{dK(t)}{dt} = s_k Y(t) - (\delta + n + g)K(t)$ $\frac{dH(t)}{dt} = s_h Y(t) - (\delta + n + g)H(t)$, where δ is the depreciation rate and g is the rate of technological progress. The Steady-State Per Capita Output y^* , which is the long-run equilibrium level of income per person, is derived when these accumulation rates are zero. The resulting steady-state level form, which serves as the core equation for the simulation analysis, is $y^* = A(0)^{\frac{1}{(1-\alpha)(1-\beta)}} \cdot s_k^{\frac{\alpha}{(1-\alpha)(1-\beta)}} \cdot s_h^{\frac{\beta}{(1-\alpha)(1-\beta)}} \cdot n^{\frac{\alpha+\beta-2\alpha\beta}{(1-\alpha)(1-\beta)^2}}$. This final equation confirms the central model predictions: the steady-state level of per capita output is positively dependent on the initial technology level $A(0)$

and the saving rates for both physical s_k and human capital s_h , while it is negatively dependent on the population or labor force growth rate n . This structured mathematical platform is essential for quantitatively assessing the structural weaknesses, such as low investment and rapid population pressures, that have historically constrained the Philippines' economic progress relative to its regional neighbors.

B. Equations and Functions Used

The foundation of this study rests upon the evolution of the Solow–Swan Neoclassical Growth Model, commencing with the basic framework used by Canlas (2003) and extending to the Extended Solow Model with Human Capital to provide a robust analytical structure for the Philippine economy.

1. Basic Solow–Swan Neoclassical Growth Model

Canlas's initial application utilized the basic Solow model, which relies on a Cobb–Douglas production function focusing on the accumulation of physical capital K and the expansion of the labor force L , while treating technology A as exogenous:

$$Y(t) = A(t)K(t)^\alpha L(t)^{1-\alpha}, \quad 0 < \alpha < 1$$

In this simplified approach, the Capital Accumulation Equation was formulated without a depreciation term ($\delta = 0$):

$$\frac{dK(t)}{dt} = sY(t)$$

where s is the gross savings rate. This structure predicts that long-run income is determined by the steady-state level of

capital per worker. The logarithmic form of the resulting Steady-State Per Capita Output y^* is:

$$\ln y(t) = \left(\frac{1}{1-\alpha}\right)[\ln A(0) + rt] + \left(\frac{\alpha}{1-\alpha}\right)\ln s - \left(\frac{\alpha}{1-\alpha}\right)\ln n$$

where r is the constant rate of technological progress, and n is the labor force growth rate. This equation confirms the basic model's core hypothesis: higher savings s positively affect y^* , while faster population growth n imposes a negative constraint.

2. Extended Solow Model with Human Capital

To address the limitations of the basic model in explaining cross-country income differences, Canlas incorporated the Extended Solow Model with Human Capital as advanced by Mankiw, Romer, and Weil (1992). This framework augments the production function by treating human capital H as a distinct, accumulating factor:

$$Y(t) = A(t)K(t)^\alpha H(t)^\beta L(t)^{1-\alpha-\beta}, \quad 0 < \alpha + \beta < 1$$

Here, β represents the income share of human capital, and the constraint $\alpha + \beta < 1$ ensures that the combined factors exhibit diminishing returns when considered separately. This extension introduces a distinct investment rate for human capital s_h , alongside the investment rate for physical capital s_k . The final derivation leads to the Extended Steady-State Per Capita Output

$y(t)$, which forms the basis for the empirical simulations in this report:

$$\begin{aligned} \ln y(t) = & \left[\frac{1-\alpha\beta}{(1-\alpha)(1-\beta)} \right] [\ln A(0) + rt] + \left(\frac{\alpha}{1-\alpha} \right) \ln s_k \\ & + \left(\frac{\beta}{1-\beta} \right) \ln s_h + \left(\frac{\alpha\beta}{1-\alpha} \right) \ln h(t) + \left(\frac{\alpha\beta}{1-\beta} \right) \ln k(t) - \left[\frac{(\alpha+\beta-2\alpha\beta)}{(1-\alpha)(1-\beta)} \right] \ln n \end{aligned}$$

This equation is crucial for the Ordinary Least Squares regression and Steady-State Simulation sections. It allows for the estimation of the separate elasticities of output per worker with respect to the accumulation of physical capital α and human capital β , providing a refined quantitative assessment of their roles relative to population pressure n in shaping the Philippines' economic trajectory.

C. Definition of Variables and Parameters

The variables and parameters central to the Extended Solow–Swan Neoclassical Growth Model with Human Capital are formally defined and classified based on their determined source within the model's system. This structure clarifies the relationship between the inputs and the long-run economic outcomes predicted for the Philippines.

Endogenous Variables:

- $Y(t)$: Total output at the time t
- $y(t)$: Output per worker, $y(t)=Y(t)/L(t)$
- $K(t)$: Stock of physical capital at time t
- $H(t)$: Stock of human capital at time t

- $k(t)$: Physical capital per worker, $k(t) = K(t)/L(t)$
- $h(t)$: Human capital per worker, $h(t) = H(t)/L(t)$

Exogenous Variables and Parameters:

- $A(t)$: Level of technology at the time t , assumed to grow at a constant exogenous rate g , so that $A(t) = A(0)e^{gt}$. In Canlas β (2003), this was represented with r , as the growth rate.
- $L(t)$: Labor force at time t , assumed to grow at a constant exogenous rate n , so that $L(t) = L(0)e^{nt}$.
- s_k : Fraction of output invested in physical capital.
- s_h : Fraction of output invested in human capital.
- α : Output elasticity of physical capital
- β : Output elasticity of human capital
- δ : Depreciation rate of capital

Empirical Proxy Variables:

- INR: Gross investment-to-GDP ratio, used as a proxy for the physical capital savings rate s_k
- NGR: Percentage change in aggregate employment, used as a proxy for the population/labor force growth rate n
- ENR: Growth rate of school enrollment (secondary or tertiary), used as a proxy for the growth of human capital per worker $h(t)$

D. Equilibrium and Steady State

The final equation from the previous section shows the general time path for the GDP per capita output. However, to obtain the actual steady-state value y^* , we set the time-dependent terms to their stationary levels. In effect, this will lead to the following: $k(t)$ converging to k^* , $h(t)$ converging to h^* , and t converging to infinity. By algebra, the level form y^* is expressed as follows:

$$\begin{aligned} \ln y^* = & \left[\frac{1-\alpha\beta}{(1-\alpha)(1-\beta)} \right] [\ln A(0) + rt] + \left(\frac{\alpha}{1-\alpha} \right) \ln s_k + \left(\frac{\beta}{1-\beta} \right) \ln s_h \\ & + \left(\frac{\alpha\beta}{1-\alpha} \right) \ln h^* + \left(\frac{\alpha\beta}{1-\beta} \right) \ln k^* - \left[\frac{(\alpha+\beta-2\alpha\beta)}{(1-\alpha)(1-\beta)} \right] \ln n \end{aligned}$$

Since $r < 0$ and $t \rightarrow \infty$, then $rt \rightarrow 0$.

$$\begin{aligned} \ln y^* = & \left[\frac{1-\alpha\beta}{(1-\alpha)(1-\beta)} \right] [\ln A(0)] + \left(\frac{\alpha}{1-\alpha} \right) \ln s_k + \left(\frac{\beta}{1-\beta} \right) \ln s_h \\ & + \left(\frac{\alpha\beta}{1-\alpha} \right) \ln h^* + \left(\frac{\alpha\beta}{1-\beta} \right) \ln k^* - \left[\frac{(\alpha+\beta-2\alpha\beta)}{(1-\alpha)(1-\beta)} \right] \ln n \end{aligned}$$

At steady state, the expressions for k^* and h^* are functions of s_k , s_h , and n , so when substituted into the equation, their log terms combine with the existing ones, allowing the entire expression to be factored and simplified into a single bracket with $\frac{1}{(1-\alpha)(1-\beta)}$.

$$\ln y^* = \left[\frac{1}{(1-\alpha)(1-\beta)} \right] [\ln A(0) + \left(\frac{\alpha}{1-\alpha} \right) \ln s_k + \left(\frac{\beta}{1-\beta} \right) \ln s_h - \frac{(\alpha+\beta-2\alpha\beta)}{(1-\alpha)(1-\beta)} \ln n]$$

By algebra, the level form y^* is expressed as follows:

$$y^* = A(0)^{\frac{1}{(1-\alpha)(1-\beta)}} \cdot s_k^{\frac{\alpha}{(1-\alpha)(1-\beta)}} \cdot s_h^{\frac{\beta}{(1-\alpha)(1-\beta)}} \cdot n^{-\frac{\alpha+\beta-2\alpha\beta}{(1-\alpha)(1-\beta)^2}}$$

III. Analysis and Results

A. Model Behavior and Dynamics

The following subsection presents empirical patterns in the model's main variables — namely, investment, employment growth, and human capital. In particular, descriptive plots of GDP per worker, investment and employment dynamics, and secondary enrollment growth serve to visually reinforce the empirical challenges and highlight the drivers underlying the Philippines' growth trajectory.



Figure 3.1. GDP per Worker from 1981 to Present

The plot illustrates the Philippines' highly volatile and often stagnant GDP per Worker trajectory, derived from the production function per capita across the entire period. This pattern directly confirms the premise of Canlas's study, that the Philippine economy was characterized by an inability to achieve sustained and rapid growth in income per person. The visual data highlights periods of severe economic crises, where sharp declines were followed by slow and often insufficient recoveries to regain pre-crisis levels quickly. This persistent volatility suggests that the fundamental Solow model drivers, specifically the accumulation

of capital (k), operated suboptimally, inhibiting the necessary capital deepening required for sustained economic expansion.

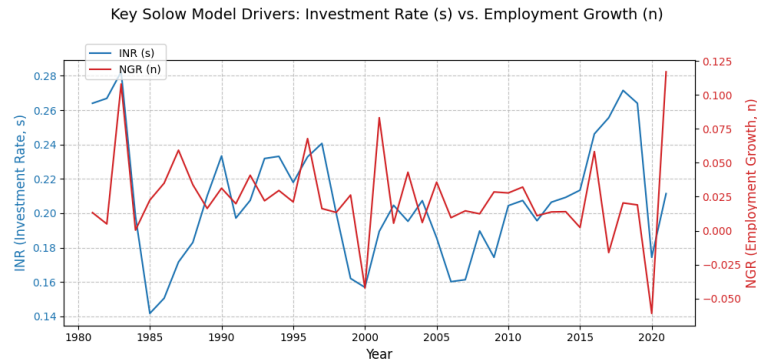


Figure 3.2. Investment Rate vs Employment Growth from 1981 to Present

The plot illustrating INR or the Investment Rate (s) and NGR or the Employment Growth (n) visually explains the core empirical results of Canlas (2003) regarding the Philippines' economic performance, the key components of the Steady-State Per Capita Output equation. The plot shows that the INR was comparatively low and highly fluctuating, supporting the finding that the coefficient for the positive effect of $\ln s$ was only weakly statistically significant; thus, the investment effort was insufficient to raise the steady-state level substantially. Conversely, the NGR is characterized by severe volatility, fluctuating with sharp upward spikes and periods of contraction, but maintaining a positive mean over the full period. This strong average rate of labor force expansion reinforces the powerful negative and statistically significant coefficient for $\ln n$, confirming the model's prediction that the consistent pressure of population growth acts as a severe economic drag by constantly forcing the limited capital to be spread among an expanding workforce.

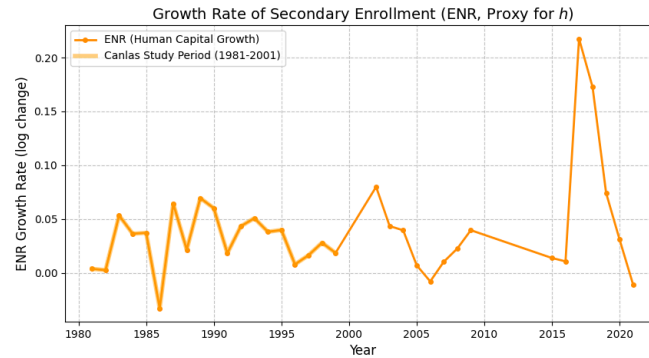


Figure 3.3. Secondary Enrollment Growth Rate from 1981 to the Present

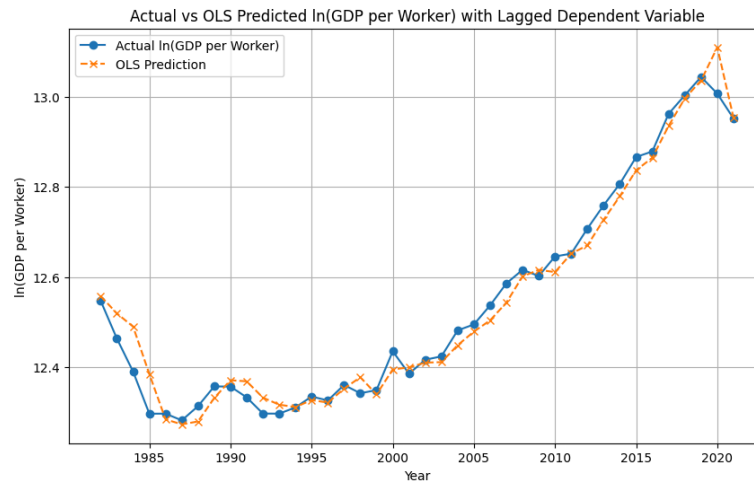
The plot of the Secondary Enrollment Growth Rate or ENR (h) illustrates the dynamics of the variable intended to capture human capital accumulation in the Extended Solow Model. The visualization reveals extreme year-to-year volatility, with the standard deviation exceeding the mean. This noise and inconsistency in the growth rate of the human capital proxy is the primary empirical reason Canlas found the corresponding coefficient to be statistically insignificant. The erratic nature of the data suggests that, while the theoretical link is valid, the proxy used was too unstable to establish a robust long-term relationship with per capita output during the study period.

B. Simulation

This section integrates the empirical findings of Canlas (2003) with a descriptive analysis and visualization of the key variables using the provided time series data from 1981 to 2021. This analysis focuses on the behavior of the output variable GDP per Worker and the proxies for the growth factors (s , n , and h).

a. Ordinary Least Squares Simulation of Canlas' Extended Solow Model

To establish an empirical baseline prior to the steady-state simulations, the simulation begins with an ordinary least squares (OLS) estimation of the reduced-form growth equations derived from the Solow framework. This step provides direct evidence on the relationship between investment, employment growth, and education proxies with output per worker, allowing comparison with the findings of Canlas. Aligning with his methodology, a lagged value of the dependent variable was also added as a regressor. By estimating these regressions over an extended time period, it can be evaluated whether the theoretical signs predicted by the model are supported in the Philippine context and highlight the statistical significance, or lack thereof, of the key explanatory variables.



OLS Regression Results						
=====						
Dep. Variable:	ln_GDPpw	R-squared:	0.977			
Model:	OLS	Adj. R-squared:	0.975			
Method:	Least Squares	F-statistic:	375.6			
Date:	Wed, 19 Nov 2025	Prob (F-statistic):	3.23e-28			
Time:	20:19:51	Log-Likelihood:	76.479			
No. Observations:	40	AIC:	-143.0			
Df Residuals:	35	BIC:	-134.5			
Df Model:	4					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.0576	0.348	-0.166	0.869	-0.764	0.649
INR	0.2745	0.217	1.264	0.215	-0.166	0.715
ENR	0.0079	0.164	0.048	0.962	-0.325	0.341
NGR	-0.7236	0.205	-3.531	0.001	-1.140	-0.308
ln_GDPpw_lag	1.0024	0.029	35.105	0.000	0.944	1.060
=====						
Omnibus:	15.960	Durbin-Watson:	1.115			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	17.838			
Skew:	-1.422	Prob(JB):	0.000134			
Kurtosis:	4.616	Cond. No.	736.			
=====						

Figure 3.4. OLS Model Plot & Summary for the Base Assumption

The OLS estimation provides a clear empirical baseline before proceeding to the steady-state simulations. The regression results confirm the expected signs from the Solow framework: the gross investment-to-GDP ratio (INR), used as a proxy for the savings rate, enters positively, while the percentage change in aggregate employment (NGR), serving as a proxy for population growth, enters negatively. Secondary enrollment growth (ENR), intended to capture human capital accumulation, also enters with the expected positive sign but is not statistically significant. Moreover, INR is only statistically significant at slightly above the 20% level. This outcome mirrors the findings of Canlas, who likewise observed that savings and employment growth were the principal drivers of per capita output, while education failed to achieve significance due to limitations in the proxy measure.

The strong coefficient on the lagged dependent variable shows that current output is closely tied to past levels, reflecting the economy's tendency to adjust

gradually over time. At the same time, the very high explanatory power of the model, with an adjusted R-squared above 0.97, indicates that the Solow framework captures most of the structural variation in Philippine output. Note that this is also noticeably higher than the 0.83-0.87 by Canlas, showing that the longer coverage enhances statistical precision without altering the core conclusions. These results establish that the theoretical predictions hold in direction, but the strength of the coefficients is uneven, particularly for INR and ENR.

b. Steady-State Simulation of Canlas' Extended Solow Model

Building on the level-form equation for y^* (GDP per capita) derived in the previous section, steady-state outcomes are simulated under alternative parameter scenarios and plotted against the observed time-series data.

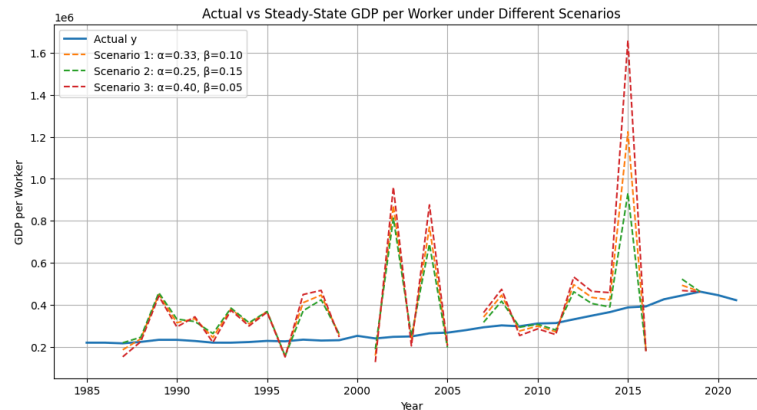


Figure 3.5. GDP per Worker under Different Capital Shares (Secondary Enrollment as Human Capital Proxy)

This simulation of the Solow model applied with human capital emphasizes both the theoretical predictions of the framework as well as the limitations of available data for the Philippines, despite drawing on a larger

dataset that spans a longer time period than that used by Canlas. Steady-state GDP per worker is highly sensitive to the assumed capital shares, with higher physical capital shares (α) amplifying the effect of investment (INR) on output, while higher human capital shares (β) increase the impact of education (ENR). These sensitivities explain why the steady-state paths differ across the scenarios, with the "middle" configuration ($\alpha = 0.25, \beta = 0.15$) producing the most plausible trajectory and tracking the actual output series more closely once extreme spikes are ignored. The model reacts sharply to sudden changes in the input proxies (most notably around 2002, 2004, and 2015) because the logarithmic transformations used in the steady-state calculations magnify relative changes in the underlying series.

IV. Evaluation of the Model

Comparisons with actual GDP per worker reveal that the observed series is smoother and much more linear than the theoretical steady-state paths. These fluctuations arise from the sensitivity of the steady-state equation to the empirical proxies used for savings (INR), human capital accumulation (ENR), and employment/population growth (NGR). Since these variables enter the model in logarithmic form, even moderate changes, particularly when ENR or INR approach zero, produce disproportionately large jumps in the theoretical steady-state series. This divergence is consistent across all α - β combinations, indicating that the volatility is driven not by parameter choice but by the limitations of the proxy variables, especially the use of ENR as a growth rate rather than a level measure of human capital. Overall, while the Solow framework

captures the qualitative direction of effects, the empirical fit remains weak due to noisy inputs and the structural complexity of the Philippine economy.

The volatility and occasional undefined values in the simulations are therefore likely not computational errors but symptoms of the mismatch between idealized theory and Philippine data. These findings mirror the empirical observation of Canlas that education has the correct sign but is not statistically significant in aggregate regressions. However, his study did not examine steady-state simulations, which distinguishes the present analysis. The simulation confirms that increases in savings and education raise the steady-state level of output, while higher population growth reduces it, yet the empirical fit remains limited by noisy or imperfect proxies. Future improvements could involve replacing ENR with enrollment ratios or average years of schooling, substituting the other variables for likewise positive versions, incorporating a low positive rate of technological progress (g) to smooth the productivity term, or applying transformations or floors to prevent negative values in the input series.

V. Recommendations

A. Proposed Improvements

To address the limitations of the standard Solow model and to extend Canlas' simulations to better reflect the current Philippine economy, several extensions are proposed that incorporate refined measures of human capital, institutional quality, and demographic structure. These modifications aim to provide a more comprehensive framework to account for key factors that interact with the country's economic growth.

Canlas' (2003) simulations included human capital, measured by secondary school enrollment rates, as a variable theoretically expected to contribute positively to economic growth. However, the results showed that this variable's effect was not statistically significant. This finding suggests that enrollment rates in the secondary level may not be sufficient to capture the true contribution of human capital. Tertiary enrollment, on the other hand, arguably provides a measure of advanced skills and specialized knowledge that directly improves the workforce quality.

Solow's basic model implicitly assumes similar institutional conditions across countries, ignoring how institutions affect investments and overall economic productivity. In an institutions-augmented Solow model by Tebaldi and Mohan (2008), they have found that the differences in the quality of institutions significantly influence both the level and growth rate of output per worker. To capture institutional quality empirically, the World Governance Indicators (WGIs) can be used as proxies, as they provide measures of the perceived quality of governance. In particular, we take the average rating for Control of Corruption, Rule of Law, and Government Effectiveness.

While the standard Solow model incorporates population growth as an exogenous factor, it does not account for variations in population quality. Since the 1980s, the Philippines has had one of the fastest population growth rates among its Asian neighbors. This resulted in a young population with high consumption, low savings, and weaker investment in physical and human capital, especially education (Canlas, 2003). The model can be extended using life

expectancy and youth dependency as proxies for demographic quality and structure. Life expectancy reflects health conditions and quality of labor, while the country's youth dependency ratio indicates potential economic burden on the workforce.

B. Simulations with Extensions of the Model

Tertiary Enrollment as the ENR Variable

OLS Regression Results						
Dep. Variable:	ln_GDPpw	R-squared:	0.979			
Model:	OLS	Adj. R-squared:	0.977			
Method:	Least Squares	F-statistic:	409.7			
Date:	Thu, 20 Nov 2025	Prob (F-statistic):	7.32e-29			
Time:	07:59:44	Log-Likelihood:	78.177			
No. Observations:	40	AIC:	-146.4			
Df Residuals:	35	BIC:	-137.9			
Df Model:	4					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.4256	0.431	0.989	0.330	-0.448	1.300
INR	0.2873	0.186	1.542	0.132	-0.091	0.665
Tertiary Enrollment (% Gross)	0.4071	0.231	1.762	0.087	-0.062	0.876
NGR (Employment Growth)	-0.7127	0.188	-3.789	0.001	-1.095	-0.331
ln_GDPpw_lag	0.9546	0.038	24.796	0.000	0.876	1.033
Omnibus:	12.015	Durbin-Watson:	1.217			
Prob(Omnibus):	0.002	Jarque-Bera (JB):	11.765			
Skew:	-1.198	Prob(JB):	0.00279			
Kurtosis:	4.146	Cond. No.	1.00e+03			

Table 5.1. OLS Result Summary with Tertiary Enrollment as Human Capital

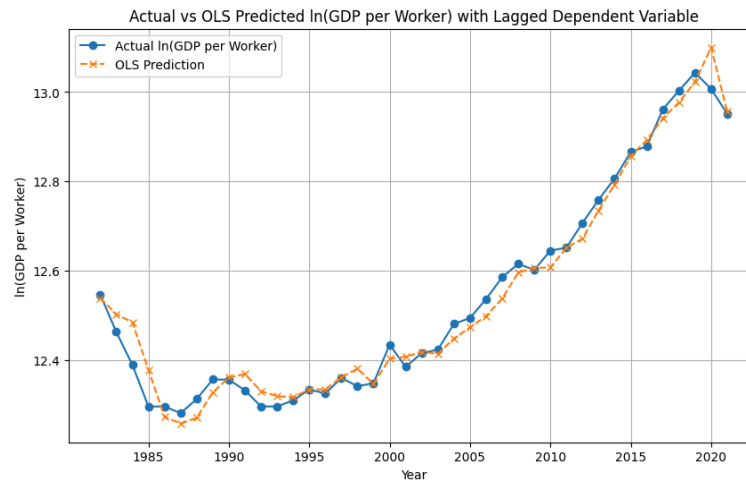


Figure 5.2. Actual vs. OLS Predicted ln (GDP per Worker)

Replacing the secondary with tertiary enrollment as the ENR proxy produces results consistent with the Solow framework while sharpening the education channel. The ENR coefficient is positive and nearly significant at the 10% level, much stronger than the 96.2% exhibited by the baseline. This could indicate that tertiary enrollment provides a more meaningful proxy for human capital accumulation than secondary enrollment, which previously failed to register. Savings (INR) remain only marginally significant, employment growth (NGR) continues to exert a strong negative effect, and the lagged dependent variable dominates, highlighting output inertia. Overall, the refinement improves education's role without altering the broader conclusion that savings and employment growth are the principal drivers of output.

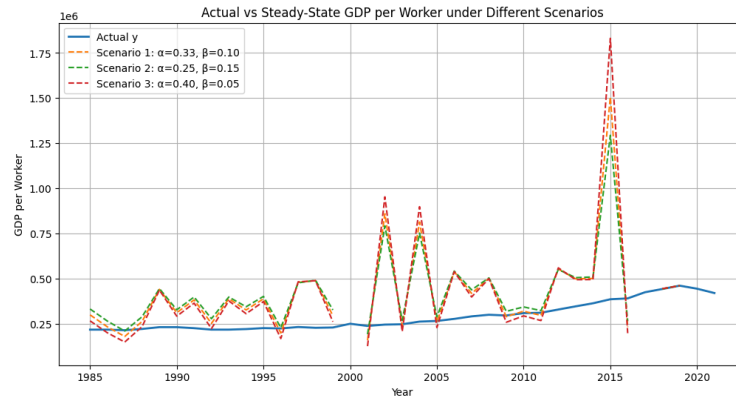


Figure 5.3 GDP per Worker under Different Capital Shares (Tertiary Enrollment as ENR)

Relative to the baseline following Canlas, the present simulation using the new ENR proxy remains consistent with the Solow framework but with some important refinements. Across all scenarios, the trajectories of GDP per worker follow the expected patterns: higher savings (INR) raise steady-state output, faster

employment growth (NGR) reduces it, and education contributes positively. The overall paths are very similar to the baseline, with both proxies generating sharp spikes in the simulated series. Thus, substituting tertiary for secondary enrollment leaves the model's core dynamics intact while offering a stronger, clearer measure of education's contribution.

Augmented Model with Institution and Demographic Structure

OLS Regression Results						
Dep. Variable:	ln_GDPpw	R-squared:	0.993			
Model:	OLS	Adj. R-squared:	0.989			
Method:	Least Squares	F-statistic:	310.1			
Date:	Thu, 20 Nov 2025	Prob (F-statistic):	7.18e-16			
Time:	09:40:00	Log-Likelihood:	60.858			
No. Observations:	24	AIC:	-105.7			
Df Residuals:	16	BIC:	-96.29			
Df Model:	7					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	7.3563	1.882	3.909	0.001	3.367	11.346
INR	0.7894	0.209	3.768	0.002	0.345	1.233
Tertiary Enrollment (% Gross)	0.2449	0.210	1.166	0.261	-0.200	0.690
NGR (Employment Growth)	-0.7749	0.196	-3.950	0.001	-1.191	-0.359
ln_GDPpw_lag	0.4008	0.152	2.638	0.018	0.079	0.723
Gov_norm	-0.0045	0.005	-0.830	0.419	-0.016	0.007
LifeExp_norm	-0.0199	0.044	-0.451	0.658	-0.113	0.074
DepYoung_norm	-0.1399	0.066	-2.105	0.051	-0.281	0.001
Omnibus:	1.930	Durbin-Watson:	1.387			
Prob(Omnibus):	0.381	Jarque-Bera (JB):	1.498			
Skew:	-0.596	Prob(JB):	0.473			
Kurtosis:	2.718	Cond. No.	5.00e+03			

Figure 5.4. OLS Result Summary with Augmented Variables

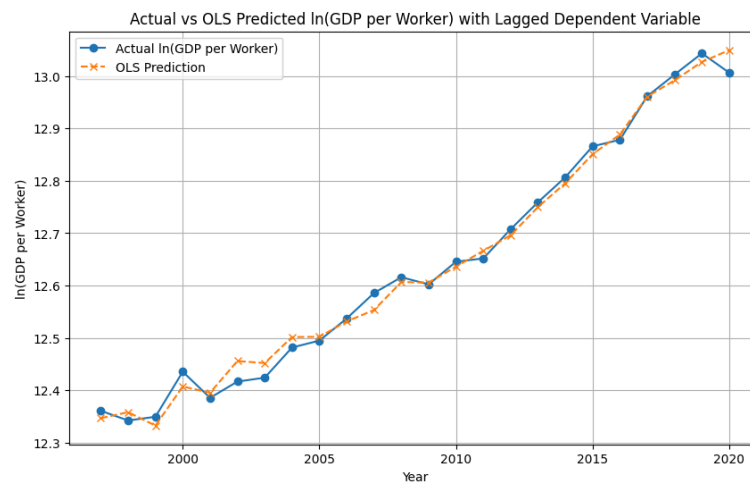


Figure 5.5. Actual vs. OLS Predicted ln(GDP per Worker) with Augmented Variables

The adjusted OLS model with the added factors, governance indicators, life expectancy, and youth dependency ratio, slightly improves the explanatory power of the regression, as shown by the increase in the adjusted R-squared from 0.977 to 0.989. Among the added variables, the youth dependency of the working-age population is the only one found to be marginally significant. Its negative coefficient suggests that a larger share of young dependents puts pressure on economic growth, as expected. While the inclusion of WGI and life expectancy in the model did not produce a statistically significant effect on GDP per worker, the model revealed a stronger impact of investment on economic growth. In Canlas' simulation, the INR is only significant at the 20% level, which was increased to 79% in this simulation.

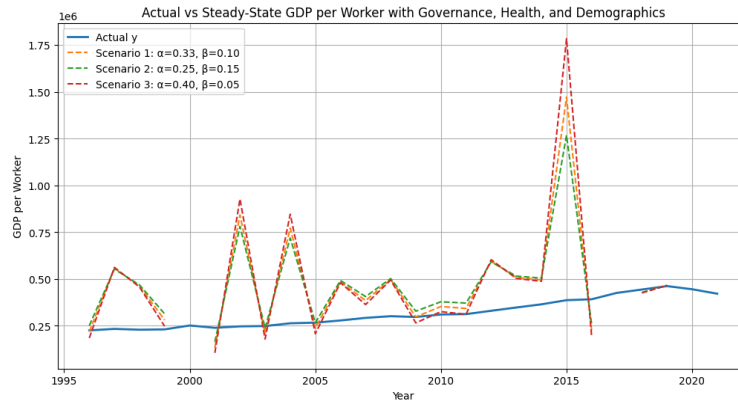


Figure 5.6. GDP per Worker under Different Capital Shares (Augmented Variables)

As a final refinement, the steady-state simulation draws on endogenous growth theory and incorporates institutional and demographic factors (WGIs, life expectancy, dependency ratio) to assess their influence on long-run output per worker. Across all scenarios simulated, the paths again behave similarly to both the baseline model of Canlas and the specification using tertiary enrollment. This

may imply that Solow's framework is quite robust, as the introduction of other variables did not produce major deviations.

VI. Conclusion

Solow's neoclassical growth model provides a useful framework for analyzing the Philippine economy. The model predicts that savings and investment positively impact economic growth, while population exerts a negative impact. Given the Philippines' low savings and investment rates alongside its rapidly growing population, the Solow framework helps explain why the country continues to lag behind its neighbors in economic growth. These predictions are strengthened by Canlas' demonstration through OLS simulations on proxy variables INR, NGR, and ENR based on an extended Solow model with human capital.

While human capital (enrollment rate in secondary level) was theoretically expected to contribute positively to economic growth, the result of Canlas' simulations shows that the variable's effect was not statistically significant. The simulations performed on an extended dataset spanning 1980-2021 also produced consistent results, with an insignificant ENR, strongly significant NGR, and INR significant only at the 20% level. On the other hand, using tertiary enrollment as a proxy for human capital resulted in ENR becoming statistically significant at the 10% level.

The problem remains that the theoretical steady-state paths are highly volatile, reflecting the sensitivity of the model to the proxy variables. To improve the reliability of future simulations, logarithmic instability may be addressed by replacing or smoothing the proxy variables for ENR, NGR, and INR to avoid extreme changes when log-transformed.

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Appendix A

Raw Data Used in Solow Model Simulations

The raw data used in this study consist of annual Philippine economic indicators. These data form the basis for all regression estimations and steady-state simulations. Due to data availability constraints, certain observations were interpolated to maintain continuity of the time series. Additionally, lagged values of GDP per Worker were generated from this raw series for simulation and regression analysis.

A1. Extended Solow Model (1981-2021)

Year	GDP per Worker	NGR	INR	ENR
1981	272605.6996	0.013265	0.264011	0.003637
1982	281302.5379	0.004924	0.266862	0.002461
1983	258675.0902	0.108103	0.282326	0.053273
1984	240375.2304	0.000378	0.198311	0.036358
1985	218967.0581	0.02248	0.141658	0.037123
1986	218987.9324	0.035008	0.150524	-0.0329
1987	215747.257	0.059294	0.1716	0.064328
1988	222668.1379	0.033806	0.183005	0.021497
1989	232639.4261	0.016327	0.210096	0.069437
1990	232541.6821	0.03126	0.233317	0.060083
1991	227023.1062	0.019838	0.197171	0.018164
1992	219030.4253	0.04082	0.207483	0.043275
1993	218993.1716	0.021993	0.231855	0.050734
1994	222004.5249	0.029579	0.233115	0.038083
1995	227464.2238	0.02114	0.217943	0.039784
1996	225491.3989	0.067865	0.232823	0.007798
1997	233393.0992	0.016252	0.240645	0.016296
1998	229120.5604	0.013411	0.199889	0.027937
1999	230738.2531	0.026219	0.162007	0.018438
2000	251498.9465	-0.04234	0.156844	
2001	239252.2929	0.08324	0.189452	
2002	246798.2556	0.005451	0.204591	0.079879
2003	248650.7599	0.04304	0.195287	0.043386
2004	263415.7054	0.005958	0.20731	0.0395
2005	266899.6078	0.035727	0.185665	0.006925
2006	278438.1185	0.009521	0.160153	-0.00801
2007	292327.1338	0.014584	0.161286	0.01022
2008	301295.9468	0.012384	0.189671	0.022508
2009	297185.848	0.028514	0.174317	0.039586
2010	310361.0655	0.02778	0.204418	
2011	312308.0338	0.032108	0.207399	
2012	330225.1646	0.01097	0.195607	
2013	347726.6285	0.013777	0.206422	
2014	364700.6997	0.013983	0.20924	
2015	386952.0005	0.002329	0.213409	0.013721
2016	391791.6946	0.058259	0.246185	0.010633
2017	425843.6576	-0.0162	0.255588	0.217628
2018	443793.0427	0.020405	0.271506	0.172677
2019	462176.3225	0.018976	0.264018	0.074247
2020	445371.6105	-0.06104	0.174334	0.030808
2021	421480.4997	0.11707	0.211407	-0.01108

A.2 Tertiary Enrollment as ENR with Sustainability & Demographic Factors (1996-2020)

Year	GDP per Worker current LCU (World Bank Data)	NGR(% Change in Aggregate Employment)	INR (Gross Investment-to-GDP ratio)	ENR (% Gross of Tertiary Enrollment)	WGI: Control of Corruption	WGI: Rule of Law	WGI: Gov't Effectiveness	Life Expectancy	Dependency Ratio, Young (% of working-age population)
1996	225491.4	0.067865	0.232823	28.52301	-0.35887	0.074511	-0.31325	66.691	72.51146
1997	233393.1	0.016252	0.240645		-0.38195	0.02726	-0.03688	66.994	71.61347
1998	229120.6	0.013411	0.199889	26.64955	-0.52759	-0.38853	-0.20413	67.212	70.39039
1999	230738.3	0.026219	0.162007	27.5218	-0.53853	-0.38654	-0.17898	67.503	69.02739
2000	251498.9	-0.04234	0.156844		-0.63537	-0.55039	-0.14702	67.789	67.76868
2001	239252.3	0.08324	0.189452	28.64768	-0.7077	-0.62895	-0.22179	67.829	66.65181
2002	246798.3	0.005451	0.204591	28.34815	-0.66284	-0.33355	-0.01642	67.886	65.66251
2003	248650.8	0.04304	0.195287	27.23011	-0.88943	-0.40689	-0.13433	68.049	64.61867
2004	263415.7	0.005958	0.20731	26.54576	-0.77801	-0.46309	0.106632	68.301	63.5589
2005	266899.6	0.035727	0.185665	25.82343	-0.7781	-0.55022	0.047636	68.423	62.52134
2006	278438.1	0.009521	0.160153	26.24681	-0.78674	-0.575	0.028276	68.172	61.64132
2007	292327.1	0.014584	0.161286		-0.78112	-0.55379	0.070901	68.549	60.80388
2008	301295.9	0.012384	0.189671	27.12899	-0.68503	-0.50639	0.189581	68.643	60.02501
2009	297185.8	0.028514	0.174317	26.60558	-0.5739	-0.52241	0.205686	68.631	59.27412
2010	310361.1	0.02778	0.204418	27.75492	-0.38007	-0.46901	0.205929	68.883	58.32141
2011	312308	0.032108	0.207399	29.07092	-0.43975	-0.36952	0.317714	69.096	57.40103
2012	330225.2	0.01097	0.195607	29.62989	-0.48113	-0.41737	0.160158	69.237	56.46512
2013	347726.6	0.013777	0.206422	31.87042	-0.51923	-0.43154	0.018497	69.283	55.52627
2014	364700.7	0.013983	0.20924	33.87419	-0.50754	-0.50129	-0.03491	69.311	54.64394
2015	386952	0.002329	0.213409	35.87459	-0.57629	-0.56514	0.062349	69.45	53.50686
2016	391791.7	0.058259	0.246185	38.34474	-0.60621	-0.5744	0.059953	69.49	52.16551
2017	425843.7	-0.0162	0.255588	33.6841	-0.51066	-0.65207	0.067812	69.966	50.88588
2018	443793	0.020405	0.271506	28.11662	-0.53011	-0.6668	0.03479	69.761	49.74378
2019	462176.3	0.018976	0.264018	30.05487	-0.54121	-0.51826	0.06393	69.68	48.70701
2020	445371.6	-0.06104	0.174334	31.63176	-0.53764	-0.41815	0.154092	70.097	47.50168

Appendix B

Python Code for Solow Model Estimation and Simulation

B.1 Code for the Extended Solow Model (Using Secondary Enrollment as ENR)

B.1.1 OLS Estimation Code

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt

df = pd.read_csv(
    "https://drive.google.com/uc?export=download&id=1-AW2LbwEiqIEFm-_2x_2IBOLbpUknPSb",
    na_values=["NA", "-", ""])
df.sort_values("Year").reset_index(drop=True)

# process missing ENR values
df['ENR'] = df['ENR'].interpolate().fillna(method='ffill').fillna(method='bfill')

df['s_k'] = df['INR']
df['s_h'] = df['ENR']
df['n'] = df['NGR']

df['ln_GDPpw'] = np.log(df['GDP per Worker'])
df['ln_GDPpw_lag'] = df['ln_GDPpw'].shift(1)
df = df.dropna(subset=['ln_GDPpw_lag']).reset_index(drop=True)

y = df['ln_GDPpw']
X = df[['INR', 'ENR', 'NGR', 'ln_GDPpw_lag']]
X = sm.add_constant(X)

model = sm.OLS(y, X).fit()
print(model.summary())
```

B.1.2 Actual vs OLS Predicted GDP per Worker Code

```
df['OLS_ln_pred'] = model.predict(X)

plt.figure(figsize=(10,6))
plt.plot(df['Year'], df['ln_GDPpw'], label='Actual ln(GDP per Worker)', marker='o')
plt.plot(df['Year'], df['OLS_ln_pred'], label='OLS Prediction', linestyle='--',
         marker='x')
plt.xlabel('Year')
plt.ylabel('ln(GDP per Worker)')
plt.title('Actual vs OLS Predicted ln(GDP per Worker) with Lagged Dependent Variable')
plt.legend()
plt.grid(True)
plt.show()
```

B.1.3 Steady-State Simulation Code under different α and β Scenarios

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(
    "https://drive.google.com/uc?export=download&id=1-AW2LbwEiqIEFm-_2x_2IBOLbpUknPSb",
    na_values=["NA", "-", ""])
df.sort_values("Year").reset_index(drop=True)

df = df[df['Year'] >= 1985].reset_index(drop=True)

# process missing ENR values
df['ENR'] = df['ENR'].interpolate().fillna(method='ffill').fillna(method='bfill')

df['s_k'] = df['INR']
df['s_h'] = df['ENR']
df['n'] = df['NGR']

delta = 0
g = 0
D = df['n'] + g + delta

# calibration base year (3rd to last since no negatives)
base_year = df['Year'].iloc[-3]
y0 = df.loc[df['Year']==base_year, 'GDP per Worker'].iloc[0]
s_k0 = df.loc[df['Year']==base_year, 's_k'].iloc[0]
s_h0 = df.loc[df['Year']==base_year, 's_h'].iloc[0]
n0 = df.loc[df['Year']==base_year, 'n'].iloc[0]
D0 = n0 + g + delta

# scenarios
scenarios = {
    "Scenario 1:  $\alpha=0.33$ ,  $\beta=0.10$ ": (0.33, 0.10),
    "Scenario 2:  $\alpha=0.25$ ,  $\beta=0.15$ ": (0.25, 0.15),
    "Scenario 3:  $\alpha=0.40$ ,  $\beta=0.05$ ": (0.40, 0.05)
}

plt.figure(figsize=(12,6))
plt.plot(df["Year"], df["GDP per Worker"], label="Actual y", linewidth=2)

for label, (alpha, beta) in scenarios.items():

    den = (1 - alpha)*(1 - beta)
    exp_sk = alpha / (1 - alpha)
    exp_sh = beta / (1 - beta)
    exp_n = (alpha + beta - 2*alpha*beta) / ((1 - alpha)*(1 - beta))

    #  $y_0 = A_0^{1/den} * s_{k0}^{exp_{sk}} * s_{h0}^{exp_{sh}} * n_0^{-exp_n}$ 
    A0_num = y0
    A0_den = (s_k0**exp_sk) * (s_h0**exp_sh) * (n0**(-exp_n))

    A0 = (A0_num / A0_den)**den

    df["y_star"] = (
```

```

        (A0**(1/den)) *
        (df["s_k"]**exp_sk) *
        (df["s_h"]**exp_sh) *
        (df["n"]**(-exp_n))
    )

    plt.plot(df["Year"], df["y_star"], linestyle='--', label=label)

plt.xlabel("Year")
plt.ylabel("GDP per Worker")
plt.title("Actual vs Steady-State GDP per Worker under Different Scenarios")
plt.legend()
plt.grid(True)
plt.show()

```

B.2 Code for the Extended Solow Model (Using Tertiary Enrollment as ENR)

B.2.1 OLS Estimation Code

```

import pandas as pd
import numpy as np
import statsmodels.api as sm
import matplotlib.pyplot as plt

df = pd.read_csv(
    "https://drive.google.com/uc?export=download&id=1kFq_r0eoGRNvzpmtsAqYKdVvi01lXAHZ",
    na_values=["NA", "-", ""])
df.sort_values("Year").reset_index(drop=True)
df = df[df['Year'].between(1981,2021)].reset_index(drop=True)

# transform to %
df['Tertiary Enrollment (% Gross)'] = df['Tertiary Enrollment (% Gross)'] / 100
# process missing ENR values
df['Tertiary Enrollment (% Gross)'] = df['Tertiary Enrollment (%
    Gross)'].interpolate().fillna(method='ffill').fillna(method='bfill')

df['s_k'] = df['INR'] # INR
df['s_h'] = df['Tertiary Enrollment (% Gross)'] # ENR
df['n'] = df['NGR (Employment Growth)'] # NGR

df['ln_GDPpw'] = np.log(df['GDP per Worker current LCU (World Bank Data)'])
df['ln_GDPpw_lag'] = df['ln_GDPpw'].shift(1)
df = df.dropna(subset=['ln_GDPpw_lag']).reset_index(drop=True)

y = df['ln_GDPpw']
X = df[['INR', 'Tertiary Enrollment (% Gross)', 'NGR (Employment Growth)',
    'ln_GDPpw_lag']]
X = sm.add_constant(X)

model = sm.OLS(y, X).fit()
print(model.summary())

```

B.2.2 Actual vs OLS Predicted GDP per Worker Code

```
df['OLS_ln_pred'] = model.predict(X)

plt.figure(figsize=(10,6))
plt.plot(df['Year'], df['ln_GDPpw'], label='Actual ln(GDP per Worker)', marker='o')
plt.plot(df['Year'], df['OLS_ln_pred'], label='OLS Prediction', linestyle='--',
         marker='x')
plt.xlabel('Year')
plt.ylabel('ln(GDP per Worker)')
plt.title('Actual vs OLS Predicted ln(GDP per Worker) with Lagged Dependent Variable')
plt.legend()
plt.grid(True)
plt.show()
```

B.2.3 Steady-State Simulation Code under different α and β Scenarios

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(
    "https://drive.google.com/uc?export=download&id=1kFq_r0eoGRNvzpmtsAqYKdVvi01lXAHZ",
    na_values=["NA", "-", ""])
df.sort_values("Year").reset_index(drop=True)
df = df[df['Year'].between(1985,2021)].reset_index(drop=True)

# transform to %
df['Tertiary Enrollment (% Gross)'] = df['Tertiary Enrollment (% Gross)'] / 100
# process missing ENR values
df['Tertiary Enrollment (% Gross)'] = df['Tertiary Enrollment (%
    Gross)'].interpolate().fillna(method='ffill').fillna(method='bfill')

df['s_k'] = df['INR'] # INR
df['s_h'] = df['Tertiary Enrollment (% Gross)'] # ENR
df['n'] = df['NGR (Employment Growth)'] # NGR

delta = 0
g = 0
D = df['n'] + g + delta

# calibration base year (3rd to last since no negatives)
base_year = df['Year'].iloc[-3]
y0 = df.loc[df['Year']==base_year, 'GDP per Worker current LCU (World Bank
    Data)'].iloc[0]
s_k0 = df.loc[df['Year']==base_year, 's_k'].iloc[0]
s_h0 = df.loc[df['Year']==base_year, 's_h'].iloc[0]
n0 = df.loc[df['Year']==base_year, 'n'].iloc[0]
D0 = n0 + g + delta

# scenarios
scenarios = {
    "Scenario 1:  $\alpha=0.33$ ,  $\beta=0.10$ ": (0.33, 0.10),
    "Scenario 2:  $\alpha=0.25$ ,  $\beta=0.15$ ": (0.25, 0.15),
    "Scenario 3:  $\alpha=0.40$ ,  $\beta=0.05$ ": (0.40, 0.05)
```

```

}

plt.figure(figsize=(12,6))
plt.plot(df["Year"], df["GDP per Worker current LCU (World Bank Data)"], label="Actual
        y", linewidth=2)

for label, (alpha, beta) in scenarios.items():

    den = (1 - alpha)*(1 - beta)
    exp_sk = alpha / (1 - alpha)
    exp_sh = beta / (1 - beta)
    exp_n = (alpha + beta - 2*alpha*beta) / ((1 - alpha)*(1 - beta))

    # y0 = A0^(1/den) * s_k0^(exp_sk) * s_h0^(exp_sh) * n0^(-exp_n)
    A0_num = y0
    A0_den = (s_k0**exp_sk) * (s_h0**exp_sh) * (n0**(-exp_n))

    A0 = (A0_num / A0_den)**den

    df["y_star"] = (
        (A0**(1/den)) *
        (df["s_k"]**exp_sk) *
        (df["s_h"]**exp_sh) *
        (df["n"]**(-exp_n))
    )

    plt.plot(df["Year"], df["y_star"], linestyle='--', label=label)

plt.xlabel("Year")
plt.ylabel("GDP per Worker")
plt.title("Actual vs Steady-State GDP per Worker under Different Scenarios")
plt.legend()
plt.grid(True)
plt.show()

```

B.3 Code for the Augmented Solow Model Incorporating Sustainability and Demographic Factors (Using Tertiary Enrollment as ENR)

B.3.1 OLS Estimation Code

```

import pandas as pd
import numpy as np
import statsmodels.api as sm
import matplotlib.pyplot as plt

df = pd.read_csv(
    "https://drive.google.com/uc?export=download&id=1kFq_r0eoGRNvzpmtsAqYKdVvi01lXAHZ",
    na_values=["NA", "-", ""])
df.sort_values("Year").reset_index(drop=True)
df = df[df['Year'].between(1996,2020)].reset_index(drop=True) # data availability
                        constraints for WIGs

# transform to %
df['Tertiary Enrollment (% Gross)'] = df['Tertiary Enrollment (% Gross)'] / 100

```

```

# process missing ENR values
df['Tertiary Enrollment (% Gross)'] = df['Tertiary Enrollment (%
    Gross)'].interpolate().fillna(method='ffill').fillna(method='bfill')

# composite governance indicator variable
df['Gov'] = df[['WGI: Control of Corruption', 'WGI: Rule of Law', 'WGI: Gov't
    Effectiveness']].mean(axis=1)

# normalize for stability
df['Gov_norm'] = (df['Gov'] - df['Gov'].mean())/df['Gov'].std()
df['LifeExp_norm'] = (df['Life Expectancy'] - df['Life Expectancy'].mean())/df['Life
    Expectancy'].std()
df['DepYoung_norm'] = (df['Dependency Ratio, Young (% of working-age population)'] -
    df['Dependency Ratio, Young (% of working-age population)'].mean())/df['Dependency
    Ratio, Young (% of working-age population)'].std()

df['ln_GDPpw'] = np.log(df['GDP per Worker current LCU (World Bank Data)'])
df['ln_GDPpw_lag'] = df['ln_GDPpw'].shift(1)
df = df.dropna(subset=['ln_GDPpw_lag']).reset_index(drop=True)

y = df['ln_GDPpw']
X = df[['INR', 'Tertiary Enrollment (% Gross)', 'NGR (Employment Growth)',
    'ln_GDPpw_lag', 'Gov_norm', 'LifeExp_norm', 'DepYoung_norm']]
X = sm.add_constant(X)

model = sm.OLS(y, X).fit()
print(model.summary())

```

B.3.2 Actual vs OLS Predicted GDP per Worker Code

```

df['OLS_ln_pred'] = model.predict(X)

plt.figure(figsize=(10,6))
plt.plot(df['Year'], df['ln_GDPpw'], label='Actual ln(GDP per Worker)', marker='o')
plt.plot(df['Year'], df['OLS_ln_pred'], label='OLS Prediction', linestyle='--',
    marker='x')
plt.xlabel('Year')
plt.ylabel('ln(GDP per Worker)')
plt.title('Actual vs OLS Predicted ln(GDP per Worker) with Lagged Dependent Variable')
plt.legend()
plt.grid(True)
plt.show()

```

B.3.3 Steady-State Simulation Code under different α and β Scenarios

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(
    "https://drive.google.com/uc?export=download&id=1kFq_r0eoGRNvzpmtsAqYKdVvi01lXAHZ",
    na_values=["NA", "-", ""])
df.sort_values("Year").reset_index(drop=True)
df = df[df['Year'].between(1996,2021)].reset_index(drop=True) # data availability
constraints for WGIs

```

```

# transform to %
df['Tertiary Enrollment (% Gross)'] = df['Tertiary Enrollment (% Gross)'] / 100
# process missing ENR values
df['Tertiary Enrollment (% Gross)'] = df['Tertiary Enrollment (%
    Gross)'].interpolate().fillna(method='ffill').fillna(method='bfill')

# composite governance indicator variable
df['Gov'] = df[['WGI: Control of Corruption', 'WGI: Rule of Law', 'WGI: Gov't
    Effectiveness']].mean(axis=1)

# normalize for stability
df['Gov_norm'] = (df['Gov'] - df['Gov'].mean()) / df['Gov'].std()
df['LifeExp_norm'] = (df['Life Expectancy'] - df['Life Expectancy'].mean()) / df['Life
    Expectancy'].std()
df['DepYoung_norm'] = (df['Dependency Ratio, Young (% of working-age population)'] -
    df['Dependency Ratio, Young (% of working-age population)'].mean()) / df['Dependency
    Ratio, Young (% of working-age population)'].std()

# endogenize
df['s_k'] = df['INR'] * (1 - 0.02*df['DepYoung_norm']) # savings penalized by youth
    dependency
df['s_h'] = df['Tertiary Enrollment (% Gross)'] * (1 + 0.01*df['LifeExp_norm']) # ENR
    boosted by life expectancy
df['n'] = df['NGR (Employment Growth)'] # NGR

delta = 0
g = 0

# governance-driven TFP shifter
df['A_t'] = np.exp(0.05*df['Gov_norm'])

# calibration base year (3rd to last since no negatives)
base_year = df['Year'].iloc[-3]
y0 = df.loc[df['Year']==base_year, 'GDP per Worker current LCU (World Bank
    Data)'].iloc[0]
s_k0 = df.loc[df['Year']==base_year, 's_k'].iloc[0]
s_h0 = df.loc[df['Year']==base_year, 's_h'].iloc[0]
n0 = df.loc[df['Year']==base_year, 'n'].iloc[0]

# scenarios
scenarios = {
    "Scenario 1:  $\alpha=0.33$ ,  $\beta=0.10$ ": (0.33, 0.10),
    "Scenario 2:  $\alpha=0.25$ ,  $\beta=0.15$ ": (0.25, 0.15),
    "Scenario 3:  $\alpha=0.40$ ,  $\beta=0.05$ ": (0.40, 0.05)
}

plt.figure(figsize=(12,6))
plt.plot(df["Year"], df["GDP per Worker current LCU (World Bank Data)"], label="Actual
    y", linewidth=2)

for label, (alpha, beta) in scenarios.items():
    den = (1 - alpha)*(1 - beta)
    exp_sk = alpha / (1 - alpha)
    exp_sh = beta / (1 - beta)
    exp_n = (alpha + beta - 2*alpha*beta) / ((1 - alpha)*(1 - beta))

```

```

# y0 = A0^(1/den) * s_k0^(exp_sk) * s_h0^(exp_sh) * n0^(-exp_n)
A0_num = y0
A0_den = (s_k0**exp_sk) * (s_h0**exp_sh) * (n0**(-exp_n))
A0 = (A0_num / A0_den)**den

df["y_star"] = (
    (A0**(1/den)) *
    (df["A_t"]**(1/den)) *
    (df["s_k"]**exp_sk) *
    (df["s_h"]**exp_sh) *
    (df["n"]**(-exp_n))
)

plt.plot(df["Year"], df["y_star"], linestyle='--', label=label)

plt.xlabel("Year")
plt.ylabel("GDP per Worker")
plt.title("Actual vs Steady-State GDP per Worker with Governance, Health, and
    Demographics")
plt.legend()
plt.grid(True)
plt.show()

```